

Anycubic Kobra 2 Neo: Phased 3D Printing & Fabrication Guide

Biomechanical Anthropomorphic Robotic Hand (Servo & Tendon Actuated)

1. Machine & Slicer Overview: Anycubic Kobra 2 Neo

- **Print Volume:** $220 \times 220 \times 250 \text{ mm}$ (Entire hand fits comfortably on this bed).
- **Extruder Type:** Direct Drive Extruder (ideal for precise retractions and optional flexible TPU finger pads).
- **Build Plate:** Magnetic Spring Steel PEI Sheet (textured).
- **Recommended Slicer:** Anycubic Slicer, PrusaSlicer, or Cura (Anycubic Kobra 2 Neo profile).

2. Global Slicer Parameters for Kobra 2 Neo

Setting	Structural Parts (PETG / PLA+)	Flexible Pads (Optional TPU 95A)
Nozzle Temperature	210°C - 215°C (PLA+) / 235°C (PETG)	220°C - 225°C
Bed Temperature	60°C (PLA+) / 75°C (PETG)	50°C
Layer Height	0.12 mm – 0.16 mm (Critical for hinge pins and tendon bores)	0.20 mm
Wall Loops / Perimeters	4 to 5 Walls (Minimum 1.6 - 2.0 mm wall thickness)	3 Walls
Top / Bottom Solid Layers	5 Top / 5 Bottom	4 Top / 4 Bottom
Infill Density & Pattern	40% – 50% Gyroid (resists multi-axial shear)	25% Gyroid
Print Speed	$60 - 80 \text{ mm/s}$ (Outer walls: 40 mm/s)	$25 - 35 \text{ mm/s}$
Retraction	$1.5 - 2.0 \text{ mm}$ @ 45 mm/s (Direct Drive)	1.0 mm @ 20 mm/s
Supports	Tree Supports (Organic) on build plate only	None / Minimal
Support Blockers	Place support blockers over $1.5 - 2.0 \text{ mm}$ internal tendon bores	N/A

3. Phased Printing Plan (Bed Batches)

To prevent failures, make post-processing easy, and maintain perfect organization, print in the following 4 chronological phases:

- Phase 1: Palm Core & Servo Adapter
- └─ palm.stl (Qty: 1)
 - └─ forearm_servo_adapter.stl (Qty: 1)

- Phase 2: Thumb Biomechanical Digits
- └─ thumb_proximal.stl (Qty: 1)
 - └─ thumb_distal.stl (Qty: 1)

- Phase 3: Primary Grasp Digits (Index & Middle)
- └─ index_proximal.stl (Qty: 1)
 - └─ index_intermediate.stl (Qty: 1)
 - └─ index_distal.stl (Qty: 1)
 - └─ middle_proximal.stl (Qty: 1)
 - └─ middle_intermediate.stl (Qty: 1)
 - └─ middle_distal.stl (Qty: 1)

- Phase 4: Ulnar Support Digits (Ring & Pinky)
- └─ ring_proximal.stl (Qty: 1)
 - └─ ring_intermediate.stl (Qty: 1)
 - └─ ring_distal.stl (Qty: 1)
 - └─ pinky_proximal.stl (Qty: 1)
 - └─ pinky_intermediate.stl (Qty: 1)
 - └─ pinky_distal.stl (Qty: 1)

Detailed Phase Specifications

Phase 1: Structural Chassis & Servo Bay

Estimated Print Time: ~5.5 - 7.5 hours

File Name	Qty	Bed Orientation	Support Required
palm.stl	1	Lay flat on dorsal back surface (wrist flange facing down or palm facing up)	Tree supports under thenar overhang & knuckle clevises
forearm_servo_adapter.stl	1	Lay vertical on flat circular base	Minimal tree supports under servo flange tabs

Pro Tip: Use a brim of 4–5 mm on the Kobra 2 Neo textured PEI sheet to prevent corners of the palm from lifting.

Phase 2: Thumb Biomechanical Digits

Estimated Print Time: ~1.5 - 2.0 hours

File Name	Qty	Bed Orientation	Support Required
thumb_proximal.stl	1	Horizontal on lateral flat edge	Tree supports under joint clevis
thumb_distal.stl	1	Upright on base clevis or angled 45°	Support under fingertip pulp pad

Verification: Test-fit a 3.0 mm dowel pin or M3 screw through the thumb hinge immediately after printing.

Phase 3: Primary Grasp Digits (Index & Middle)

Estimated Print Time: ~3.0 - 4.0 hours

File Name	Qty	Bed Orientation	Support Required
index_proximal.stl	1	Lateral side down on bed	Tree supports under female knuckle clevis
index_intermediate.stl	1	Lateral side down on bed	Tree supports under male clevis tongue
index_distal.stl	1	Angled 45° or upright	Support under tactile pad
middle_proximal.stl	1	Lateral side down on bed	Tree supports under female knuckle clevis
middle_intermediate.stl	1	Lateral side down on bed	Tree supports under male clevis tongue
middle_distal.stl	1	Angled 45° or upright	Support under tactile pad

Phase 4: Ulnar Support Digits (Ring & Pinky)

Estimated Print Time: ~2.5 - 3.5 hours

File Name	Qty	Bed Orientation	Support Required
ring_proximal.stl	1	Lateral side down on bed	Tree supports under clevis
ring_intermediate.stl	1	Lateral side down on bed	Tree supports under clevis
ring_distal.stl	1	Angled 45° or upright	Support under tactile pad
pinky_proximal.stl	1	Lateral side down on bed	Tree supports under clevis
pinky_intermediate.stl	1	Lateral side down on bed	Tree supports under clevis
pinky_distal.stl	1	Angled 45° or upright	Support under tactile pad

4. Master Parts Inventory & Quantity Check

Part Category	STL File Name	Required Quantity	Cumulative Total
Chassis	palm.stl	1	1
Forearm	forearm_servo_adapter.stl	1	2
Thumb	thumb_proximal.stl	1	3
Thumb	thumb_distal.stl	1	4
Index	index_proximal.stl	1	5
Index	index_intermediate.stl	1	6
Index	index_distal.stl	1	7
Middle	middle_proximal.stl	1	8
Middle	middle_intermediate.stl	1	9
Middle	middle_distal.stl	1	10
Ring	ring_proximal.stl	1	11
Ring	ring_intermediate.stl	1	12
Ring	ring_distal.stl	1	13
Pinky	pinky_proximal.stl	1	14
Pinky	pinky_intermediate.stl	1	15
Pinky	pinky_distal.stl	1	16
Reference	robotic_hand_full_assembly.stl	(CAD reference only)	—
Reference	thumb_assembly.stl	(CAD reference only)	—

Total printable structural parts: 16 individual pieces.

5. Post-Processing & Calibration for Anycubic Kobra 2 Neo

1. Pin Hole Post-Processing:
- 3D printer shrinkage typically tightens small holes by 0.1 - 0.2 mm.

◦ Take a 3.0 mm drill bit and ream every hinge pin hole by hand (do not use high-speed power drills which can melt the polymer).
2. Tendon Bore Clearing:
- Run a 1.2 mm - 1.5 mm guitar string (G or D string) or rigid floral wire through every internal cable routing channel to ensure zero burrs or snag points.
3. Clevis Dressing:
- Use 400-grit wet sandpaper on the sides of the male hinge tongues for friction-free movement.
4. Bed Leveling (Kobra 2 Neo LeviQ 2.0):
- Run the automatic LeviQ 2.0 bed leveling calibration before Phase 1 to guarantee uniform first-layer adhesion across the entire palm surface.